Expert Upcycling: Shifting the Compute-Efficient Frontier of Mixture-of-Experts

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Research Questions

- How can we progressively expand MoE capacity during training while preserving per-token inference cost and achieving comparable quality to training from scratch?
- What theoretical framework can decompose the quality gap between upcycled and fixed-size MoE models to guide optimization?
- How can gradient-based importance scores guide non-uniform expert duplication to improve quality gap closure in expert upcycling?

Core Idea

Training large MoE models from scratch is expensive due to memory and communication costs scaling with total parameters. Expert upcycling starts with a smaller E-expert model, duplicates experts at transition step τ to create mE experts while keeping top-K routing fixed, then continues training. The key insight is that duplicated experts provide "warm initialization" - inheriting learned representations instead of random weights - and subsequent training breaks symmetry through stochastic gradients to drive specialization. This saves compute since early training uses the smaller model while achieving comparable final quality.

Key Formula

Expert Upcycling Bound

$$L^{up} - L^{fs} \leq \frac{\sum_{t=0}^{\tau-1} \eta_t}{\sum_{t=0}^{T-1} \eta_t} (L_E^\star - L_{mE}^\star) + \frac{\|\theta_+^U - \theta_+^\star\|^2 - \|\theta_+^0 - \theta_+^\star\|^2}{2 \sum_{t=0}^{T-1} \eta_t}$$

Decomposes quality gap between upcycled (L^up) and fixed-size (L^fs) into capacity term (time in smaller model class) and initialization term (how well upcycling operator θ_+^U initializes new parameters vs random θ_+^0). η_t is learning rate, θ_+^* is optimal new parameters.

Key Figure

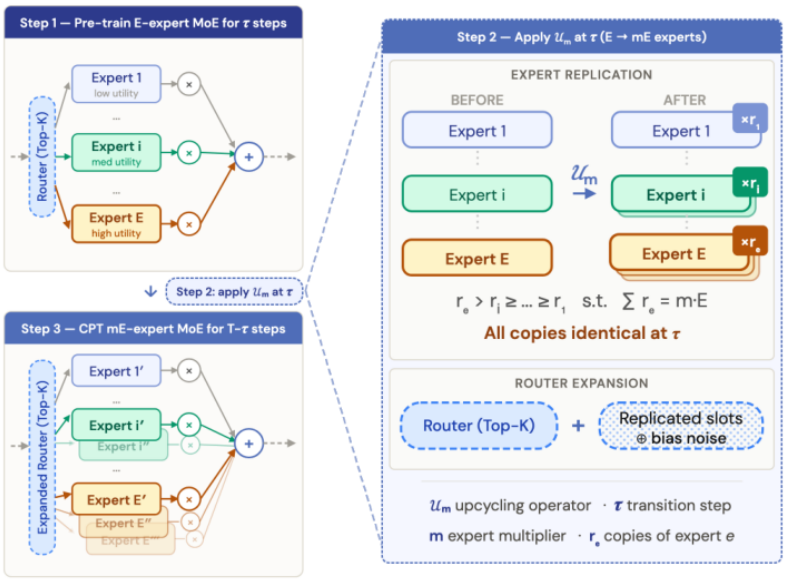

Figure 1: **Overview of the expert upcycling procedure. Step 1:** Pre-train an E-expert MoE for τ steps. **Step 2:** Apply the upcycling operator $\mathcal{U}_m$ at step τ: each expert e is replicated $r_e \geq 1$ times (high-utility experts receive more copies, $r_E > r_i \geq \cdots \geq r_1$, s.t. $\sum r_e = m \cdot E$), and the router is extended with replicated slots plus bias noise. All copies are identical at τ, providing a warm initialization. **Step 3:** Continue pre-training on the expanded mE-expert model for $T-\tau$ steps; stochastic gradient diversity breaks symmetry among duplicates, driving specialization. Top-K routing is fixed throughout, so active parameters and per-token compute are unchanged.

Figure 1: Illustrates the complete expert upcycling procedure showing how experts are replicated with utility-based selection and router extension while maintaining identical per-token computation.

Limitations

- Only validated with m=2 expert doubling; larger expansion factors may face new challenges
- Experiments limited to 7B scale; frontier model scaling may reveal router collapse or balancing brittleness
- Distribution shift between pre-training and CPT phases not thoroughly explored
- Theoretical framework relies on convexity assumptions that don't hold for deep networks